

[11] Power Supply

No communication is needed between nPM1300 and the USB host. The CC lines can be used to detect the USB host max current capability, but even then it is not automatic, you can read the detected value in the register and SoC can set the VBUSINLIM accordingly through TWI. So you can just set the VBUS current limit to the desired value with TWI write when the 5V adapter is plugged. Note the default is 100mA, so that is the maximum you can draw from the 5V supply with the default setting. You can program it up to 1500mA.

nPM1300 settings:

- VOUT1: 2.7v
- VSET1: 300k
- VOUT2: 3.3V
- VSET2: 300k

Vout1 can only be 2.7V top trough VSET, but we can set trough sw the vout and it can be 3.3V

each of the bucks (vout1 and vout2) can provide up to 200mA, not more.
The vout1 will be 1v8 and will be used for the nrf7000, (which should have a maximum current consumption of 64mA in pasive mode) in active mode it consumes around 250mA on peak). We need to take that into account to stick with pasive.

NOTE: C55 and C56 should be placed close to PVSS1 and PVSS2 pads respectively.

VSYS: 2500mA max
Connected to:
-nr9151

Initially its 2v7. But it will be changed trough SW to 3v3

VOUT1:
120mA Max current from Lora
45mA from BMV080 Analog
22.4mA from BMV080 Digital
Total: 190mA

Buck 2: 200mA max
Connected to:
-IMU
-Flash
-GPIOs
-nr54lm20

A maximum of 50mA can be drawn x LDO

NPM1300

Vibrator power switch

Speaker power switch

Comments:

Company:

Variant:

Git Hash:

Board Name:

aeris_pcb

Project Name:

Palroc One

Sheet Title:

Power Supply

File Name:

PowerSupply.kicad_sch

Designer:

Oriol Reverte

Date:

Revision:

Sheet Path:

/Project Architecture/Power Supply/

Reviewer:

Size:

A3

Sheet:

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